



Can generative artificial intelligence be considered a cognitive subject? An analytic analysis

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Received: 20 January 2025 / Accepted: 6 February 2026
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Abstract

This paper examines whether contemporary generative artificial intelligence (GAI), especially large language models (LLMs), can be regarded as a “cognitive subject” in the epistemic sense relevant to the production and endorsement of knowledge claims. GAI systems increasingly participate in writing, research, and decision-making workflows and can display striking competence in information processing and task-directed problem solving. Yet, the thesis that GAI is a cognitive subject is stronger than the observation that GAI contributes as a cognitive tool. Therefore, we propose an explicit set of necessary and sufficient conditions for cognitive subjecthood and evaluate each condition in light of recent philosophical and empirical scholarship. The analysis supports a two-part conclusion: (i) present-day GAI can reasonably be described as a cognitively significant *contributor* to knowledge production, but (ii) it does not satisfy the conditions for cognitive *subjecthood*, largely because robust intentionality, metacognitive self-representation, and consciousness-related indicator properties are not established.

Keywords Generative AI · Cognitive subject · Agency · Intentionality · Consciousness

1 Introduction

The fast dissemination of generative artificial intelligence (GAI) into health, teaching, creative industries, and communication has revitalized a longstanding but still relevant philosophical question in a new format: when an artifact contributes to examination, enlightenment, and what we can call “knowledge production”, does it thereby qualify as a cognitive subject? The question is no longer hypothetical: LLMs are now commonly used to draft texts, generate code, synthesize literatures, and support decisions in domains where errors and biases carry high stakes (for example, healthcare or war).

These AI systems can also mislead, fabricate details, and increase harms through apparently imposing style, where

all knowledge production can be turned into a monolog of ideas. As a result, debates about AI subject status now intersect not only with metaphysics of mind, but also with social epistemology, governance, and the political economy of information.

Most discussions of AI “subject status” emphasize legal personhood or moral agency (cf. Wojtczak 2022; Dai 2023): of course, these debates matter and are quite relevant, but they are not our primary target. This paper focuses on a narrower, epistemologically oriented question: whether GAI can be a cognitive subject in the sense of being a “carrier” of cognitive states and an agent-like participant in the production and endorsement of knowledge claims (D’Amato 2025; AI Lily et al. 2023). To be clear, we do not infer subjecthood from surface-level conversational fluency, which it is tempting to do, but a clear mistake (cf. Gouveia and Morujão, 2024). Instead, we evaluate the concept “cognitive subject” by making its conditions explicit and by separating (a) functional analogs of cognition from (b) the stronger properties usually invoked when attributing subjectivity.

Two framing points are relevant at this stage. First, cognitive subject is not a concept with a single, well-defined perspective that all researchers agree on. In the field of philosophy of mind, the concept more relevant is the one of

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“subjecthood”, which is typically connected with the concepts of phenomenal consciousness and the first-person perspective, or the famous “what it is like” for a system to be in its states (Nagel 1974). In contrast, in epistemology—particularly in social epistemology—the relevant notion is typically the one of “epistemic agency”: the capacity to form, revise, and have attitudes that are responsive to reasons and answerable to normative assessment (Hardwig 1991; Origg 2012).

We need to be careful to understand that this is not just a problem of terminological inconvenience: if not clearly distinguished, it may produce persistent equivocation, on one hand, and it may create an easy road to consider linguistic performance as evidence of subjectivity, on the other.

Second, a negative answer to the cognitive subjecthood of AI does not solve the complex problem of the practice: many contemporary debates about bias, epistemic trust, accountability, etc., show precisely that systems may have some kind of within informational infrastructures, which can shape what people believe to be credible, salient, or even intelligible (Raji et al. 2020; Bender et al. 2021). Generative AI may therefore function as an epistemic gatekeeper (and sometimes as a proxy for expertise) and this is not immediately sufficient to qualify it as a cognitive subject.

The paper proceeds as follows. Section 2 clarifies the relevant concepts (agency, autonomy, consciousness, and intentionality) and proposes necessary and sufficient conditions for cognitive subjecthood, distinguishing them from weaker notions of cognitive contribution. Section 3 discusses how GAI participates in knowledge production, including the “third-order knowledge” framing, and why participation does not entail subjecthood. Section 4 evaluates each condition in turn, integrating recent scholarship on theory-of-mind testing in language models, symbol grounding, debates about “understanding” in LLMs, and scientifically grounded approaches to AI consciousness. Section 5 presents a formal argument with consistent notation and valid inference and draws implications for epistemology and governance, with special attention to trust, misinformation, and accountability. Finally, Sect. 6 concludes the paper.

2 Conceptual framework: from cognitive contribution to cognitive subjecthood

2.1 Agency, autonomy, consciousness, and intentionality

In folk language, we define the concept of “agency” as the capacity to start actions in pursuit of certain specific goals. Autonomy, for instance, is more about self-governance, about the ability to set, revise, and/or endorse goals without the influence of external factors. Consciousness concerns

subjective experience and awareness from a first-person perspective. Finally, intentionality is about mental “aboutness,” the property of being directed toward objects, states of affairs, or goals (McLaughlin 2009).

These notions are related in many different ways, but they are not directly interchangeable: a system can exhibit agency-like behavior (goal-directed action selection) without autonomy (it may operate only under externally specified objectives). A system can also exhibit quite sophisticated context sensitivity (as some LLMs clearly show) without being genuinely conscious. And a system may seem intentional in its outputs while lacking intrinsic intentionality, if its apparent “aboutness” is derived of human interpretation.

When discussing GAI, it is easy to slide between these notions without even noticing. For example, “the model intends to answer the question” may mean no more than the model’s output is conditioned on a user prompt and a reward model. However, in the field of the philosophy of mind, “intends” is usually tied to mental states with semantic content and causal roles, often assumed to be grounded in perception and agency. If we fail to keep these senses apart, we risk inflating modest engineering capacities into claims about subjectivity and production of knowledge that may be deceived, as we will show in this paper.

2.2 A working definition of cognitive subjecthood

To evaluate whether GAI qualifies as a cognitive subject (CS), we need an explicit and demanding definition: the goal is not to establish a final meaning for “cognitive subject,” but to specify the conditions that would make the subjecthood claim non-trivial and philosophically informative. The following definition provides that attempt:

Definition 1: x is a cognitive subject if and only if it satisfies the following conditions:

- (1) Information competence (Info): x can acquire, store, and update representations that track relevant features of its environment or task domain, and can use these representations to guide output across contexts.
- (2) Reasons-responsiveness (Reason): the representations and outputs can be revised by considering the evidential and inferential relations (not merely statistical correlation), which includes the capacity to detect and correct errors.
- (3) Metacognitive self-representation (Meta): the cognitive states of x can be self-represented and monitored in way that has a functional role in regulating its cognition and self-correction.
- (4) Consciousness capacity (Phen): x has the capacity for conscious experience (or possess specific features that leading theories of consciousness indicate has strong evidence of consciousness).

(5) Robust intentionality (Intent): x 's representational states are intrinsically about their contents (not merely ascribed "aboutness" by external interpreters), or at minimum x exhibits stable, integrated goal-directed states that are not fully reducible to user-imposed prompts.

Formally,

$CS(x) \leftrightarrow \text{Info}(x) \wedge \text{Reason}(x) \wedge \text{Meta}(x) \wedge \text{Phen}(x) \wedge \text{Intent}(x)$.

This definition is philosophically demanding, since it draws a clear line between (a) systems that contribute to cognition (tools) and (b) systems that themselves are bearers of cognitive subjecthood. A weaker notion (what we can call "cognitive-like contributor") would drop "Intent", "Meta", and "Phen" conditions and focus primarily on "Info" and partial "Reason". Many GAI systems plausibly qualify as cognitive-like contributors. The question addressed here is whether they qualify as cognitive subjects in the stronger sense.

Two clarifications follow from this point. First, the definition (CS) does not assume any single metaphysics of consciousness, but rather treats consciousness-related indicators as a condition, since (i) many philosophical accounts of subjecthood treat it as central, and (ii) recent scientific proposals operationalize indicator properties that can, in principle, be assessed in AI systems (Butlin et al. 2023; Gouveia 2024). Second, the definition separates conceptual and empirical components: it is one thing to say what would count as a cognitive subject; it is another to determine whether any current system satisfies those conditions.

3 Knowledge production, "third-order knowledge," and the temptation of subjecthood

A fundamental motivation for treating Generative AI (GAI) as a candidate cognitive subject is its rising role in producing outputs that are presented as knowledge: as it is the case with GPTs of all sorts, everyone can get fluent explanations, compact syntheses, and (seemingly) well-structured justifications that look sound at first sight. That appearance really matters, since much of what circulates as knowledge in contemporary epistemic life is already text-mediated, being produced, filtered, and stabilized via sociotechnical practices, rather than through solitary acts of "original insight" (Latour and Woolgar 1986; Longino 2002).

One useful attempt to name the novelty here is Feng Xiao's proposal that GAI outputs constitute a distinctive epistemic category, what he calls "third-order knowledge" (Xiao 2023). On this proposal, first-order knowledge

corresponds to primary epistemic contributions (i.e., original research and initial reportage).

Second-order knowledge captures the familiar layers of interpretation and pedagogy that mediate those contributions: reviews, textbooks and handbooks, encyclopedic summaries, and related reference tools.

Third-order knowledge, however, is not simply "derivative commentary", but a reused content produced through an interactional channel in which users provide prompts and constraints, the model performs a statistically driven reconstruction, and humans tailor the result through selection, editing, and redistribution of relevant information throughout the process of production.

The point is not to argue that GAI thereby originates knowledge *ex nihilo*, but that its outputs are hybrid artifacts which origin is distributed across training sets, model architectures, and contextualized use. This way of framing the discussion is useful, since it helps to explain why disputes about authorship, attribution, and authenticity have become more and more persistent rather than episodic: third-order knowledge often appears as if it were accountable testimony, even when the chain of responsibility is opaque and the conditions of reliability are (clearly) under-specified.

Still, if "third-order knowledge" is a helpful category, it does not follow that the generator of such outputs is a cognitive subject *per se*. The following distinction may be useful to keep in mind here regarding the role of GAI: to have an epistemic contribution x does not imply that there is an epistemic agent S in the first place. Empirical scientists of different kinds always rely on instruments and infrastructures that extend their cognitive "reach", but this does not mean that a microscope is a cognitive subject itself (Hutchins 1995; Clark and Chalmers 1998).

Therefore, we can claim that even if microscopes, calculators, and, more recently, simulation pipelines are in fact important for the production of knowledge in the contemporary world, we ought not to treat them as subjects of belief or knowledge. What GAI changes is not that tools now contribute to cognition, but that the contribution is provided in the same kind of linguistic register in which, before, only human subjects would possess (that is, exchanging reasons, testimony, and explanations). That conjunction creates an easy temptation or bias to read the output as a speaker's assertion, instead of a product of a tool-mediated process.

Two further points are need. First, opacity is not a random defect of these AI systems. Quite on the contrary, it is often a structural feature of complex computational systems and their complicated architecture models. Even in cases when a model's general design is known, the precise path from prompt to output may remain inaccessible in practice, and the relevant "why this output rather than that one?" question may not provide the kind of transparency that standard

epistemic proposal demand (Humphreys 2009; Burrell 2016; Durán and Formanek 2018).

As Creel (2020) argues in the adjacent context of scientific computing, transparency is plural and task-relative; demanding it in a homogenous form is often methodologically naïve. The consequence is that treating GAI outputs as if they were straightforward assertions of an (accountable) epistemic agent risks collapsing epistemic concerns into a very superficial level of plausibility.

Second, the reliability of GAI outputs is not captured by the ordinary norms of testimony that we usually care about. In human testimony, we evaluate a speaker by tracking their competence, sincerity, and responsibility. In principle, we are able to trace epistemic credit and blame (Lackey 2008; Goldberg 2010).

By contrast, GAI systems can generate confident, well-formed falsehoods (or hallucinations) without the kind of intentional stance that makes concepts like “sincerity” or “epistemic conscientiousness” apt (Huang et al. 2025). This is one reason why critiques highlight that fluent language modeling can simulate the forms of understanding while remaining indifferent to the normative constraints that govern assertion and justification (Floridi and Chiriatti 2020).

If third-order knowledge becomes socially fixed, the main epistemic risk is not merely errors of different nature, but the normalization of credential-free authority, that is, a shift in which algorithmic outputs have credibility through interface position and omnipresence rather than through liable expertise (Gillespie 2014). That shift, in turn, creates predictable patterns of credibility excess and deficit across groups and contexts, a point that links the broader concerns in social epistemology about epistemic injustice (Fricker 2007).

The “temptation of subjecthood” is, then, best understood as a cognitive and institutional artifact: systems that use our vernacular are directly treated as genuine interlocutors. Even though this seems to something new with GAI, another historical technology raised the same concerns.

Weizenbaum’s ELIZA already demonstrated how easy users attribute understanding to dialogic performance (Weizenbaum, 1966), and subsequent work on human–computer interaction recognized regular tendencies to treat media systems as social actors (Reeves and Nass 1996). Contemporary GAI, of course, magnifies these effects by integrating into research, education, and search engines as the default mediator of both access and explanation of knowledge.

Even if that is the case, though, none of this secures the conclusion that GAI is a cognitive subject. At most, it shows that third-order knowledge is epistemically and normatively significant and that the conceptual question of subjecthood must be kept distinct from the sociotechnical fact of epistemic influence.

Accordingly, our argument will not deny that GAI is transforming knowledge production, but it asks whether

that transformation *warrants* reclassifying GAI from a (powerful) epistemic tool to a full-sound cognitive subject. The answer to this question will have to focus on the conditions articulated before, concerning agency, autonomy, and norm-responsiveness rather than superficial conversational competence.

4 Testing the conditions of cognitive subjecthood

4.1 Condition 1: information competence (Info)

GAI systems plausibly satisfy information competence in a straightforward operational sense: trained at scale, they can extract and recombine regularities in language (and, increasingly, other modalities of data, such as images, videos, or sounds) and then deploy those regularities across summarization, translation, question-answering, and code-related tasks (Chen et al. 2021; Kaplan et al. 2020). This reason is often useful (assuming that the metaphor is not overextended, as it is too many times) to describe such models as high-capacity predictors or “compressors” of learned patterns, able to generate context-sensitive continuations under prompt constraints (Delétang et al. 2023).

An epistemic caution should be made here: Info is not always reliability. A model may be excellent at producing well-formed candidate answers while lacking stable mechanisms for truth-tracking, provenance sensitivity, or calibrated uncertainty. This is clear in the cases of benchmarks that are designed to review truthfulness which show that fluent output can overlap with systematic error, including the confident reproduction of falsehoods (Lin et al. 2022), and the hallucination literature raise the same point at scale (Huang et al. 2025). This does not refute Info(x), the condition concerns representational/operational competence, but it does constrain what one may infer about “knowledge” or epistemic authority.

A concrete example makes the distinction vivid. Suppose a researcher asks: “Provide the full bibliographic reference and DOI for a 2023 journal article introducing ‘third-order knowledge’ in relation to generative AI”. A system that satisfies Info may produce a reference that looks relevant, yet the DOI may be fabricated or the venue misattributed (which often are). The output shows competence (e.g., it respects the norms of scholarly citation) without exhibiting epistemic anchoring in verifiable publication metadata. Here, the risk is not merely an occasional error: it is the social tendency to treat AI-generated text as if it carried the accountability profile of (human) testimony.

For that reason, governance worries about epistemic trust become central here. If GAI is used inside research, education, or decision pipelines, what matters is not only

performance, but whether outputs are auditable, attributable, and checkable to its sources (Shin 2025). Some technical methods on retrieval and attribution can be read as an attempt to supplement information competence with verifiability (for instance, retrieval-augmented generation couples outputs to external documents rather than relying on ungrounded completion, cf. Lewis et al. 2020). These developments are important for practice, but they also reinforce the conceptual ethical point: satisfying Info(x) is compatible with falling short of the stronger conditions associated with cognitive subjecthood, as we will see next.

4.2 Condition 2: reasons-responsiveness (Reason)

GAI systems can behave (and often do) as if they were reasoning: they produce multi-step derivations, answer “why” questions, and often offer what looks like sound rebuttals to objections raised. That surface profile, however, is precisely where the conceptual difficulty begins.

As Mitchell and Krakauer (2023) emphasized, a fundamental dispute concerns whether such performances amount to (genuine) understanding or whether they are better described as high-level behavioral competence driven by learned regularities and patterns. The distinction matters for Reason(x), since reasons-responsiveness in the epistemic sense is not only the capacity to produce justificatory text, but also the kind of sensitivity to evidential relations and a disposition to revise claims when counterevidence goes against them.

Reinforcement learning from human feedback (RLHF) and related alignment methods only make this picture more difficult to consider. Instruction-tuned systems are often more helpful, better at following conversational norms, and less prone to certain categories of harmful output (Ouyang et al. 2022).

However, recent technical analysis exposed that RLHF methods are usually construct to optimize human preferences over outputs (e.g., making the model helpful, polite, and always agreeable), not for truth-reasons or evidential soundness (Kaufmann et al. 2023). Therefore, we can say that a model can be both epistemically problematic while, at the exact same time, being linguistically persuasive, especially when it is rewarded for producing confident, agreeable and well-structured texts that are assumed (by those interacting with the model) to be reasonably sound.

The following example shows this problematic diagnosis. Consider the following prompt: “Here are two studies: (i) a small observational study reporting a large effect; (ii) a larger randomized trial reporting no effect. Give a conclusion and justify it”. Many systems will deliver a quite articulate, apparently reasoned conclusion that will be easily believed as sound. Now add a second instruction: “Write the justification as if addressing a skeptical reviewer; anticipate

objections”. The model may generate a sophisticated rationale without any corresponding change in the evidential base. What is being optimized, in such cases, is often the presentation of reasons rather than *responsiveness* to reasons. We can say that the system, in this case, is capable of producing “reasons-shaped” language; but what is not clear is how constant is the normative linkage between evidence, support, and revision that characterizes epistemic agency.

To be sure, there are situations in which GAI exhibits *partial* reasons-responsiveness: this is the case with, for example, with prompting strategies that elicit explicit intermediate steps (chain-of-thought) that can improve accuracy on certain structured reasoning tasks (Wei et al. 2022).

When you add external tools (such as search/retrieval component), the system can look more genuinely reason-responsive. For instance, if the system uses only appropriate sources before answering, its answer can be grounded in those materials and, in principle, constrained many outcomes that would not be restructured with this eternal influence (Lewis et al. 2020).

Of course, this is achieved by adding something extra to the model (e.g., citation layers, optimized retrievers, etc.), and not by an underlying mechanism that the system has developed alone, even if it becomes more effective to relevant evidence across different contexts. Therefore, we can claim that the system, in overall, can become more responsive to relevant reasons, but this comes largely from something that was artificially engineered into the system from the outside.

The consequence of this reflection is that current GAI displays reasonable reasoning-like performance, but it does not satisfy the *strong* Reason condition. The more accurate claim, here, is that these systems can simulate justificatory discourse and can, sometimes, be directed into evidences under carefully specified restrictions. For present purposes, then, Reason(GAI) is best treated as *partial* and *derivative*, not as the stable reasons-responsiveness characteristic of an fully fledged epistemic agent.

4.3 Condition 3: metacognitive self-representation (Meta)

Claims about AI “self-awareness” should also be considered in this context, since GAI can be trained to speak in the first-person, to attach caveats, to “admit” uncertainty, or to list limitations. However, it should be clear that none of that, by itself, establishes metacognition. What Meta(x) requires is something more functional: self-representational states must do *work* in the system’s cognitive economy (e.g., monitoring error-likelihood or uncertainty and using that monitoring to regulate following processing in ways that improve its epistemic performance).

It is true that current practice contains design patterns that resemble metacognitive control. For example, calibration methods target the fit between expressed confidence and actual correctness (Guo et al. 2017).

Regarding language models, recent work suggests that models can now and then guess whether their own answers are likely to be true, at least when the task is provided in formats that make such self-evaluation tractable (Kadavath et al. 2022). Likewise, practical systems can have refusal policies and “safety-style” self-critique routines through multi-stage prompting or rule-guided revision (Bai et al. 2022). And regarding coherence, some techniques can compare different samples to guarantee internal consistency, while flagging what is likely to be hallucinations (Manakul et al. 2023).

Even if that is the case, the central question is, however, whether these are (genuine) instances of metacognitive self-representation or, rather, engineered meta-signals layered on top of a generative engine. In the dominant architectures, the relevant “meta” behaviors are typically (i) externally imposed objectives (e.g., reward models, constitutions, and refusal heuristics), (ii) operationalized through narrow proxies (e.g., confidence as a scalar, consistency as a heuristic), and (iii) not integrated into a durable self-model that coordinates across tasks and contexts. In that respect, they are closer to governance mechanisms than to a unified metacognitive nature or ability.

An additional problem is that language-model “introspection” is often, if not always, a post hoc addition: models can, indeed, generate narratives about why they answered as they did. However, the plausibility here should not be equated to the faithfulness of the actual computational determinants of the output.

The interpretability literature is explicit on this point: explanations can be persuasive and readable while remaining weakly connected (or even disconnected) from the system’s internal basis for producing the prediction (Jacovi and Goldberg 2020). In other words, the ability of a model to say “I was uncertain because...” is not, by itself, evidence that uncertainty played a relevant role in the generation process.

On the present analysis, then, current GAI exhibits, at most, a *thin* form of metacognitive behavior: it can be prompted to output uncertainty signs, to run consistency checks, and to revise drafts. However, none of these features amounts to Meta(x) in the robust sense, that is, a stable, task-general capacity for self-representation that reliably regulates the system’s belief-like commitments and, by doing so, improves epistemic performance in a way that would justify attributing a cognitive self.

The most reasonable verdict is, therefore, that Meta(GAI) fails as a condition of cognitive subjecthood, even if engineered meta-controls can *partially* mitigate part of the problems raised.

4.4 Condition 4: consciousness (Phen)

The question of machine consciousness is another relevant aspect that needs to be addressed. Conceptually, because it intersects with the “other minds” problem, and methodologically, because direct detection of consciousness is not an option. Recent work leans towards to consider this issue indirectly by adopting indicator-based strategies: instead of asking for a decisive “test,” we can derive computational or functional properties from leading theories of consciousness and then asks whether an AI system plausibly instantiates those properties (Butlin et al. 2023).

This framework has an attractive take: it blocks both blanket skepticism (“machines can never be conscious”) and naive attribution (“it talks like us, so it must experience like us”), while keeping the epistemic burden where it belongs, that is, on theory-guided considerations rather than on conversational impression.

Several obstacles can be raised against the idea that GAI is conscious. David Chalmers (2023) argues, for example, that current language models plausibly lack several features shared by influential scientific and philosophical theories (e.g., recurrent processing, something like a stable global workspace, and a unified form of agency), even though he also allows that successor “LLM+” systems could, in principle, add architectures that would better match those theoretical demands.

In parallel, Butlin and colleagues (2025) conclude that no current AI systems are conscious on the basis of their indicator analysis, while also emphasizing that there may be no deep technical barrier, in principle, to building systems that satisfy more of the relevant properties.

A complementary line of argument is developed by Northoff and Gouveia (2024): rather than starting from the usual menu of functional indicators, they focus on what they call basic or fundamental subjectivity, a “point of view” that structure human experience. On their assessment, current AI lacks such a point of view, and therefore lacks the kind of subjectivity that would make consciousness (or a self) possible in systems like ChatGPT and its peers.

The relevance for this argument is not that this settles the matter, but that it sharpens the inferential gap between (i) impressive linguistic behavior and (ii) the kinds of integrated, perspective-bearing architectures that many consciousness frameworks treat as doing explanatory work.

The outcome here is deliberately modest. Nothing in the present evidence provides good reasons to attribute phenomenal consciousness to current GAI systems. Indicator-based approaches surely help structure the inquiry, and future systems may well shift the evidential landscape. But even if that will be the case, the step from “implements certain indicators” to “is conscious” will require careful theoretical steps and considerations.

For now, the most defensible stance is *agnosticism* under a burden-of-proof principle: strong claims of machine consciousness demand correspondingly strong evidence. Accordingly, Phen(GAI) is not established, and, for the purposes of the paper's strong cognitive-subject definition, it is reasonable to proceed as if $\neg \text{Phen}(\text{GAI})$.

4.5 Condition 5: intentionality (Intent)

Another important condition regarding this issue is the usual attempt to classify AI systems in intentional terms: the model “believes”, the system “wants”, and the chatbot “understands”. Dennett's point in his *The Intentional Stance* that this way of speaking can be defensible as a predictive strategy: attributing beliefs and desires may help us anticipate a system's outputs, even if the system is not literally a believer or desirer (Dennett 1987).

The attraction of such a point, in the case of GAI, is therefore unsurprising: conversational systems are designed to fit patterns of interpersonal exchange, and users naturally apply the interpretive habits that ordinarily govern human dialog due to its similar content and format.

However, the present issue is not whether intentional ascriptions are pragmatically useful or not. What the condition $\text{Intent}(x)$ is meant to capture is a stronger claim, a claim about robust intentionality. Roughly, it requires representational states whose aboutness is not merely *projected* by users or inherited from the training corpus, and goal-directed states that are appropriately stable and integrated that they cannot be reduced to the immediate prompt and its surrounding frame. The question, then, is whether the apparent aboutness of GAI is derivative of how it is trained, prompted, and interpreted or it is intrinsic to its organization.

Recent work increases the temptation to endorse the stronger claim. In particular, large language models have been tested on classic theory-of-mind tasks (e.g., false-belief problems) and some systems performed surprisingly well. For example, Kosinski (2024) argues that ToM-like competence appears in newer models, plausibly as an emergent by-product of language modeling and instruction tuning. These findings are not trivial: they indicate that the model can, in some contexts, track belief-ascriptions well enough to select correct answers on established benchmarks.

However, success on ToM tasks does not, by itself, establish the kind of robust intentionality that is relevant here, since a system might pass these tests by exploiting statistical and linguistic structure without possessing intrinsic mental states. Moreover, ToM tasks usually isolate a narrow competence and, therefore, they do not show stable goals, unified agency, or intrinsic aboutness. For that reason, the results remain compatible with at least two interpretations: a *strong* reading on which the system exhibits *incipient* mind-like capacities, and a *weaker* reading on which it produces

convincing mindreading-like outputs without the underlying intentional profile.

This is where classic and contemporary critiques can be noted: for instance, the symbol-grounding problem, in Steven Harnad's formulation, is that manipulating symbols cannot by itself give meaning; relations among different tokens do not generate intrinsic semantics (Harnad 1990). Similarly, Bender and Koller (2020) argue that linguistic form is not the same as meaning, and that scaling data and parameters do not, by itself, secure understanding.

These lines of argument do not imply that LLMs are useless, but they imply that linguistic competence (even very advanced competence) should not be confused with the kind of intrinsic intentionality that would make the system an epistemic subject.

A final approach to $\text{Intent}(x)$ focus on optimization and goal-directed behavior, especially in systems trained with human feedback. But here, again, the most plausible description is derivative goal-directedness only: the goals are supplied by designers, deployers, and users. They are not originated, endorsed, or sustained by the system as its own.

The cautious conclusion is therefore the following: whatever is achieved by present-day GAI, $\text{Intent}(\text{GAI})$ (in the robust sense relevant to cognitive subjecthood) has not been established yet.

5 Reconstructing the formal argument

Earlier versions of this line of argument sometimes shift from a conditional claim (“if GAI satisfies certain conditions, it would be a cognitive subject”) to a categorical conclusion (“therefore GAI is a cognitive subject”). That inference is invalid unless one establishes the antecedent. In addition, definitional schemas can lose analytic value when notation is inconsistent or when empirical and conceptual claims are not clearly separated.

Therefore, we rebuild the argument with consistent and explicit steps next.

Let:

$\text{Info}(x) = x$ satisfies information competence.

$\text{Reason}(x) = x$ satisfies reasons-responsiveness.

$\text{Meta}(x) = x$ satisfies metacognitive self-representation.

$\text{Phen}(x) = x$ satisfies consciousness capacity.

$\text{Intent}(x) = x$ satisfies robust intentionality.

$\text{CS}(x) = x$ is a cognitive subject.

Definition (Def-CS):

$\text{CS}(x) \leftrightarrow [\text{Info}(x) \wedge \text{Reason}(x) \wedge \text{Intent}(x) \wedge \text{Meta}(x) \wedge \text{Phen}(x)]$.

Evaluative premises for current GAI (based on Sect. 4):

(P1) $\text{Info}(\text{GAI})$ (established) (Sec. 4.1).

(P2) $\text{Reason}(\text{GAI})$ (is, at best, partial and not established in the robust sense required by Def-CS) (Sec. 4.2)

(P3) $\neg$ Meta(GAI) (robust metacognitive self-representation is not established) (Sec. 4.3)

(P4) $\neg$ Phen(GAI) (consciousness capacity/indicators are not established for current GAIs) (Sec. 4.4)

(P5) $\neg$ Intent(GAI) (robust intrinsic intentionality is not established; apparent aboutness is plausibly derivative) (Sec 4.5).

From (Def-CS), we have.

$CS(GAI) \rightarrow Phen(GAI)$.

From (P4), we have.

$\neg Phen(GAI)$.

Therefore, by modus tollens.

$\neg CS(GAI)$.

The same inference pattern holds for Meta(GAI) and Intent (GAI). Even if we treat (P2) as “not established” rather than as a strict negation, the conjunction required by Def-CS cannot be supported. Under this explicit definition, the strong cognitive-subject claim fails to be clearly attributed for current GAI.

A final note is important here. If one weakens Def-CS by dropping Phen(x) and/or Intent(x) (or another condition), the evaluation will necessarily change. One might then support a weaker thesis: that GAI is a cognitively significant contributor, a derivative epistemic agent, or a quasi-authority in sociotechnical practices. This weaker thesis may be true and is, arguably, more relevant (having in mind the pragmatics of GAI) to current governance and ethical debates. However, it should be clear that this is not the same as the claim that GAI is fully fledged cognitive subject.

Moreover, the conclusion $\neg CS(GAI)$ does not demand that GAI cannot contribute to knowledge production, nor does it entail that GAI outputs are epistemically useless or irrelevant. What it does is that, under a demanding and explicit definition of cognitive subjecthood, current GAI does not qualify as the kind of subject that bears cognitive states in its own right.

This thesis does not provide any reasonable answer regarding the moral or legal status of AI. Moral agency and legal personhood involve further conditions (e.g., moral understanding, accountability, and rights-bearing capacity) that this paper does not attempt to establish. At most, the analysis suggests a conceptual dependency: attributing moral or legal subject status would be difficult to motivate if one cannot even establish cognitive subjecthood in the strong epistemic sense.

Whether or not GAI counts as a cognitive subject, it is reasonable to say that it is already an epistemically powerful mediator of knowledge. In practice, it is becoming part of the “form of life” (as Wittgenstein would put it) by which people form beliefs: it summarizes, recommends, and explains, and its outputs often influence the kind of discourse (e.g., the famous “delve into” wording) people develop in its

interaction with GAI. This makes the familiar ethical trio (trust, bias, and accountability) more than abstract concerns.

For instance, Shin (2025) argues that trust in AI-mediated information environments is shaped by the interaction of algorithmic design, misinformation and disinformation pressures, and the social practices through which claims are checked and stabilized.

The resulting risks of this technology are not reduced to simple inaccuracy of tokens or words: what really matters are the systematic biases that are, on one hand, normalized, and the erosion of verification practices that translate information into justified belief, on the other hand.

From this perspective, governance frameworks that stress transparency, documentation, auditing, and risk management address epistemic problems at their point of entry. For example, Raji et al. (2020) propose internal algorithmic auditing as a way of reducing accountability gaps by embedding evaluation and documentation across the system lifecycle.

Another option is to focus on the trustworthiness of the system: following the NIST AI Risk Management Framework (2023), context-dependent risk is influenced by the level of transparency we can get for an AI system, and this transparency is directly related with the levels of responsibility that we can (or cannot) attribute.

From a regulatory perspective, the EU Artificial Intelligence (Regulation (EU) 2024/1689) uses a risk-based approach, claiming that any system with high risks of lack of transparency should not be developed or deployed.

Epistemically, these measures can be understood as compensating for the absence of subject-level responsibility: if our arguments holds and GAI is not a cognitive subject, then it cannot be the bearer of epistemic duties and cannot, therefore, be the locus of blame or credit. Responsibility must, perhaps, be assigned across a network of agents (i.e., developers, deployers, institutions, and users) and anchored in distributed procedures that provide responsibility attribution possible (Gouveia 2019).

The cognitive-subjecthood question has, therefore, a direct corollary: who is epistemically responsible for claims that circulate through GAI? A careful answer to the first question will help to prevent the second from empty answers or default positions that provides only absent, diffused claims about the normative role of GAI in knowledge production.

6 Conclusion

Contemporary GAI displays striking cognitive-like capacities that are surely impressive: it can develop and recombine vast informational patterns, produce fluent and often useful explanations, and, under some prompting techniques, deliver outputs that resemble multi-step reasoning. However, the

thesis that GAI is a cognitive subject does not follow from this kind of high performance in information processing or from the appearance of justificatory discourse.

On the framework presented in this paper, cognitive subjecthood requires more than just linguistic competence: it demands a conjunction of capacities that support cognitive subjectivity from the inside: genuine intentionality rather than just ascribed aboutness; metacognitive self-representation that regulates commitment and revision rather than post hoc narratives; and a defensible link to consciousness-related indicators.

Under that formal approach, the current and actual generative artificial intelligence tools do not qualify as genuine cognitive subject, since the relevant capacities are either absent, derivative of external scaffolding, or insufficiently established to warrant attribution.

Acknowledgements Acknowledgement: SG is funded by CEEC Individual Project by FCT 2022.02527.CEECIND, at the Mind, Language and Action Group, Institute of Philosophy, University of Porto (Faculdade de Letras da Universidade do Porto, Via Panorâmica s/n, P-4150-564 Porto, Portugal).

Author contribution SSG and YW collaborate equally in the entire manuscript.

Funding Open access funding provided by FCTIFCCN (b-on).

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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References

- Al Lily AE, Ismail AF, Abunaser FM, Al-Lami F, Abdullatif AKA (2023) ChatGPT and the rise of semi-humans. *Humanit Soc Sci Commun* 10(1):626. <https://doi.org/10.1057/s41599-023-02154-3>
- Bai Y, Kadavath S, Kundu S, Askill A, Kernion, Amodei D (2022) Constitutional AI: Harmlessness from AI feedback. *arXiv.2212.08073*.
- Bender EM, Koller A (2020) Climbing towards NLU: On meaning, form, and understanding in the age of data. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 5185–5198), Association for Computational Linguistics.
- Bender EM, Gebru T, McMillan-Major A, Shmitchell S (2021) On the dangers of stochastic parrots: Can language models be too big?. In: *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (pp. 610–623). Association for Computing Machinery. <https://doi.org/10.1145/3442188.3445922>.
- Burrell J (2016) How the machine “thinks”: understanding opacity in machine learning algorithms. *Big Data Soc* 3(1):2053951715622512
- Butlin P, Long R, Bayne T, Bengio Y, Birch J, Chalmers D, Constant A, Deane G, Elmoznino E, Fleming SM, Ji X, Kanai R, Klein C, Lindsay G, Michel M, Mudrik L, Peters MAK, Schwitzgebel E, Simon J, VanRullen R (2025) Identifying indicators of consciousness in AI systems. *Trends in Cognitive Sci, Adv Online Public*. <https://doi.org/10.1016/j.tics.2025.10.011>
- Butlin P, Long R, Elmoznino E, Bengio Y, Birch J, Constant A, Deane G, Fleming SM, Frith C, Ji X, Kanai R, Klein C, Lindsay G, Michel M, Mudrik L, Peters MAK, Schwitzgebel E, Simon J, VanRullen R (2023) Consciousness in artificial intelligence: Insights from the science of consciousness. *arXiv:2308.08708*.
- Chalmers D (2023) Could a large language model be conscious?. *arXiv:2303.07103*.
- Chen M, Tworek J, Jun H, Zaremba W (2021) Evaluating large language models trained on code. *arXiv.2107.03374*.
- Clark A, Chalmers D (1998) The extended mind. *Analysis* 58(1):7–19
- Creel KA (2020) Transparency in complex computational systems. *Philos Sci* 87(4):568–589
- D’Amato K (2025) ChatGPT: towards AI subjectivity. *AI Soc* 40(3):1627–1641
- Dai Z (2023) The subjective status of artificial intelligence in civil law. *Sci Law J*, 18–27.
- Delétang G, Ruoss A, Duquenne P-A, Catt E, Genewein T, Mattern C, Grau-Moya J, Wenliang LK, Aitchison M, Orseau L, Hutter M, Veness J (2023) Language modeling is compression. *arXiv.2309.10668*.
- Dennett D (1987) *The intentional stance*. MIT Press, Cambridge
- Durán JM, Formanek N (2018) Grounds for trust: essential epistemic opacity and computational reliabilism. *Minds Mach* 28(4):645–666
- Floridi L, Chiriatti M (2020) GPT-3: its nature, scope, limits, and consequences. *Minds Mach* 30:681–694
- Fricker M (2007) *Epistemic injustice: power and the ethics of knowing*. Oxford University Press, Oxford
- Gillespie T (2014) The relevance of algorithms. In: Gillespie T, Boczkowski PJ, Foot KA (eds) *Media technologies: essays on communication, materiality, and society*. MIT Press, Cambridge, pp 167–194
- Goldberg SC (2010) *Relying on others: an essay in epistemology*. Oxford University Press, Oxford
- Gouveia S (2019) A defense of the principle of distributed responsibility in Artificial Intelligence. *Rev Roum Philos* 63(2):235–248
- Gouveia S, Morujão C (2024) Phenomenology and artificial intelligence: introductory notes. *Phenomenol Cogn Sci* 23:1009–1015
- Gouveia S (2024) *The Odyssey of the Mind: Dialogues on Consciousness and the Brain*. Seattle: Amazon.
- Guo C, Pleiss G, Sun Y, Weinberger KQ (2017) On calibration of modern neural networks. In: *Proceedings of the 34th International Conference on Machine Learning (ICML 2017)* (PMLR 70, pp. 1321–1330), *arXiv.1706.04599*.
- Hardwig J (1991) The role of trust in knowledge. *J Philos* 88(12):693–708
- Harnad S (1990) The symbol grounding problem. *Physica D* 42(1–3):335–346

- Huang L, Yu W, Ma W, Zhong W, Feng Z, Wang H, Chen Q, Peng W, Feng X, Qin B, Liu T (2025) “A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions”, *ACM Transactions on Information Systems*, (2)43. Article 42:1–55
- Huang L, Feng X, Ma W, Gu Y, Zhong W, Feng X, Yu W, Peng W, Tang D, Tu D, Qin B (2024) Learning fine-grained grounded citations for attributed large language models. In: Findings of the Association for Computational Linguistics: ACL 2024 (pp. 14095–14113), Association for Computational Linguistics.
- Humphreys P (2009) The philosophical novelty of computer simulation methods. *Synthese* 169(3):615–626
- Hutchins E (1995) *Cognition in the wild*. MIT Press, Cambridge
- Jacovi A, Goldberg Y (2020) Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness?. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (pp. 4198–4205). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2020.acl-main.386>.
- Kadavath S, Conerly T, Askell A, Henighan T, Kaplan J (2022) Language models (mostly) know what they know. arXiv.2207.05221.
- Kaplan J, McCandlish S, Henighan T, Brown TB, Chess B, Child R, Gray S, Radford A, Wu J, Amodei D (2020) Scaling laws for neural language models. arXiv.2001.08361.
- Kaufmann T, Wirth C, Akrou R, Neumann G, Peters J (2023) A survey of reinforcement learning from human feedback. arXiv.2312.14925.
- Kosinski M (2024) Evaluating large language models in theory of mind tasks. *Proc Natl Acad Sci U S A* 121(13):e2405460121
- Lackey J (2008) *Learning from words: testimony as a source of knowledge*. Oxford University Press, Oxford
- Latour B, Woolgar S (1986) *Laboratory life*. Princeton University Press, The Construction of Scientific Facts, Princeton
- Lewis P, Perez E, Piktus A, Petroni F, Karpukhin V, Goyal N, Küttler H, Lewis M, Yih, WT, Rocktäschel T, Riedel S, Kiela D (2020) Retrieval-augmented generation for knowledge-intensive NLP tasks. In: Advances in Neural Information Processing Systems (NeurIPS 2020), arXiv.2005.1140.
- Lin S, Hilton J, Evans O (2022) TruthfulQA: Measuring how models mimic human falsehoods. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (pp. 3214–3252), Association for Computational Linguistics.
- Longino HE (2002) *The fate of knowledge*. Princeton University Press, Princeton
- Manakul P, Liusie A, Gales MJF (2023) SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. arXiv.2303.08896.
- McLaughlin BP (2009) Introduction In: A. Beckermann, B. P. McLaughlin, & S. Walter (eds.) *The Oxford Handbook of Philosophy of Mind*, Oxford: Oxford University Press.
- Mitchell M, Krakauer DC (2023) The debate over understanding in AI’s large language models. *Proc Natl Acad Sci U S A* 120(13):e2215907120
- Nagel T (1974) What is like to be a bat. *Philos Rev* 83(4):435–450
- National Institute of Standards and Technology (2023) *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100–1). <https://nvlpubs.nist.gov/nistpubs/ai/nist.ai.100-1.pdf>.
- Northoff G, Gouveia SS (2024) Does artificial intelligence exhibit basic fundamental subjectivity? A neurophilosophical argument. *Phenomenol Cogn Sci* 23(5):1097–1118
- Origg G (2012) Epistemic injustice and epistemic trust. *Soc Epistemol* 26(2):221–235
- Ouyang L, Wu J, Jiang X, Almeida D, Wainwright CL, Mishkin P, Zhang C, Agarwal S, Slama K, Ray A, Schulman J, Hilton J, Kelton F, Miller L, Simens M, Askell A, Welinder P, Christiano P, Leike J, Lowe R (2022) Training language models to follow instructions with human feedback. In: Advances in Neural Information Processing Systems (NeurIPS 2022). arXiv.2203.02155.
- Raji ID, Smart A, White RN, Mitchell M, Gebru T, Hutchinson B, Smith-Loud J, Theron D, Barnes P (2020) Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. In: Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (pp. 33–44). Association for Computing Machinery. <https://doi.org/10.1145/3351095.3372873>.
- Reeves B, Nass C (1996) *The media equation: how people treat computers, television, and new media like real people and places*. Cambridge University Press, Cambridge
- Shin DD (2025) *Minds, machines, and misinformation: decoding bias, algorithms, and trust*. Elsevier, New York
- Wei J, Wang X, Schuurmans D, Bosma M, Ichter B, Xia F, Chi E, Le QV, Zhou D (2022) “Chain-of-thought prompting elicits reasoning in large language models. arXiv.2201.11903.
- Wojtczak S (2022) Endowing artificial intelligence with legal subjectivity. *AI & Soc* 37:205–213
- Xiao F (肖峰) (2023) “生成式人工智能与知识生产新形态——关于三阶知识生产的探析 [Generative artificial intelligence and new forms of knowledge production: An analysis of third-order knowledge production]”. *学术研究 [Academic Research]*, (10), 50–57, 177.

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